

Finite-Jet Contraction and the All-Order Obstruction

Liang Wang

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Abstract

The trace-adaptive selector of RH-238 enforces a noise-dependent finite-jet tolerance [2]. We prove the exact consequence. If $J_{m,R}(\tau_\sigma) \leq \varepsilon_\sigma$ and $\varepsilon_\sigma \rightarrow 0$, then the residual jets are Cauchy in the $J_{m,R}$ seminorm; in fact

$$J_{m,R}(\tau_\sigma - \tau_{\sigma'}) \leq \varepsilon_\sigma + \varepsilon_{\sigma'}.$$

All 30 adjacent-scale and 16 dual-channel archived cases satisfy the corresponding triangle bound. The minimum slacks are 0.00246 and 0.00220. The actual adjacent distances are not monotonically contracting, and between one and ten shell prefixes can meet a given tolerance. Most importantly, fixed- m Cauchy convergence does not control coefficients above order m . Thus adaptive finite jets advance the coefficient route but do not imply a locally uniform regularized determinant limit.

1 Finite-jet topology

For trace vectors through order m , define

$$J_{m,R}(\tau) = \sum_{n=2}^m \frac{|\tau_n|}{n} R^n.$$

This is a seminorm because the first coordinate is ignored. It is the natural coefficient majorant for the truncated $\det_2$ logarithm.

Theorem 1.1 (Tolerance contraction). *Suppose a family τ_α satisfies $J_{m,R}(\tau_\alpha) \leq \varepsilon_\alpha$. Then*

$$J_{m,R}(\tau_\alpha - \tau_\beta) \leq \varepsilon_\alpha + \varepsilon_\beta.$$

If $\varepsilon_\alpha \rightarrow 0$, the family converges to zero in the fixed finite-jet seminorm.

Proof. The first inequality is the triangle inequality. Taking α, β in a tail where both tolerances are small proves the Cauchy property. Taking the assumed one-point bounds directly gives convergence to the zero jet. $\square$

Remark 1.2. The zero limit reflects the normalization chosen in RH-238: the adaptive cloud is selected to make the residual finite jet small. It is not yet the coefficient anchor required to identify a nontrivial deterministic numerator.

Corollary 1.3 (Coefficientwise finite-order control). *If $R > 0$ and $J_{m,R}(\tau_\alpha) \leq \varepsilon_\alpha$, then for every $2 \leq n \leq m$,*

$$|\tau_{\alpha,n}| \leq \frac{n\varepsilon_\alpha}{R^n}, \quad |\tau_{\alpha,n} - \tau_{\beta,n}| \leq \frac{n(\varepsilon_\alpha + \varepsilon_\beta)}{R^n}.$$

Proof. Each nonnegative weighted coefficient is bounded by the full defining sum; apply the same observation to Theorem 1.1. $\square$

Thus vanishing tolerances give simultaneous convergence of every coefficient in the fixed finite window, with explicit rates inherited from the tolerance schedule.

2 Archived bounds

For adjacent scales the theorem gives $J(\tau_{\sigma_i} - \tau_{\sigma_{i+1}}) \leq \sigma_i + \sigma_{i+1}$. For two channels at the same scale it gives the bound 2σ .

Quantity	value
Adjacent cases	30
Channel cases	16
Minimum adjacent bound slack	0.0024624
Minimum channel bound slack	0.0022040
Maximum actual adjacent distance	0.0231244
Both last-four actual sequences strictly contract	no
Admissible-prefix count range	1–10

Table 1: Finite-jet contraction audit.

The deterministic upper bounds decrease with the noise schedule, but the actual distances fluctuate because the selected rank and shell phases jump. This distinction matters: a theorem can provide Cauchy control even when a short finite sequence is not pointwise monotone.

The last four left-channel distances happen to contract strictly, while the right-channel sequence does not. Neither observation changes the theorem: monotonicity is not required for Cauchy convergence, and a short monotone tail would not prove the all-order determinant statement.

3 Why finite jets are insufficient

For each fixed m , convergence of the first m coefficients gives only convergence of a Taylor polynomial. A sequence of entire functions can have all first m coefficients equal and still grow arbitrarily at order $m + 1$. For example,

$$f_k(z) = \exp(kz^{m+1})$$

has the same logarithmic jet through order m for every k , but is not locally bounded on any disk containing a nonzero point with positive real z^{m+1} .

Therefore the implication

$$\text{vanishing fixed finite jet} \implies \text{normal entire family}$$

is false. Uniformity in the order is indispensable.

Theorem 3.1 (No finite-jet normality criterion). *Fix $m \geq 2$, a nonzero point z_0 , and $K > 0$. There is an entire zero-free function F_K , normalized by $F_K(0) = 1$, whose logarithmic coefficients through order m all vanish, but $|F_K(z_0)| = e^K$. Consequently no bound involving only a fixed logarithmic m -jet can imply local boundedness on a neighborhood containing z_0 .*

Proof. Take

$$F_K(z) = \exp\left(K(z/z_0)^{m+1}\right).$$

Its normalized logarithm begins at order $m + 1$, while $F_K(z_0) = e^K$. Letting $K \rightarrow \infty$ violates local boundedness. $\square$

The theorem is deliberately formulated for zero-free entire functions, so the obstruction persists inside the natural class of regularized determinant quotients; it is not caused by inserting arbitrary poles or zeros.

4 Two legitimate continuations

There are two ways to strengthen the topology. The direct route controls all orders at once by a geometric envelope $|\tau_n| \leq Mq^n$. A second possible route lets the audited order $m = m(\sigma)$ grow, but then it must also provide a uniform tail estimate beyond $m(\sigma)$. Merely increasing the computed order without such a tail bound repeats Theorem 3.1 at a later index.

Both routes must be combined with an anchor. Convergence of the adaptive jets to zero proves that the chosen quotient tends toward one in every fixed audited coefficient; it does not prove that one is the intended deterministic relative factor.

5 Next wall

RH-240 supplies the missing sufficient form: a geometric envelope $|\tau_n(\sigma)| \leq Mq^n$ for every $n \geq 2$ [1, 3]. The archived orders 2–12 fit such an envelope with $q < 1$, but order thirteen begins the unaudited tail. In parallel, an anchoring theorem must show that the cloud has not absorbed the regular deterministic numerator.

No Gate A closure or zeta-zero statement is made.

References

- [1] Barry Simon. *Trace Ideals and Their Applications*. American Mathematical Society, second edition, 2005.
- [2] Liang Wang. Trace-adaptive shell-complete cloud selection. 2026. RH-238 preprint.
- [3] Liang Wang. A uniform trace-envelope criterion for relative regularized determinants. 2026. RH-240 preprint.